Computer Vision and PID Control Algorithms for Autonomous Lane Following and Obstacle Avoidance

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***Abstract*—Autonomous navigation in low-light environments remains a challenging task, particularly when relying on computer vision-based techniques. This work presents an autonomous lane following and obstacle avoidance system developed in the Webots simulator, using the Tesla Model 3 in the "City Night" environment. The system integrates computer vision techniques with a PID (Proportional–Integral–Derivative) controller to ensure real time performance and robustness. The lane detection pipeline uses OpenCV for HSV (Hue, Saturation, Value) color filtering to isolate yellow lane markings, followed by Canny edge detection and Hough Line Transform to estimate lane geometry. Steering is controlled via a PID controller using a combined lane error, derived from both the line angle and its position relative to the image center, while obstacle avoidance relies on LiDAR (Light Detection And Ranging) data to dynamically adjust heading and velocity. Experiments were conducted on two road maps with varying camera resolutions and PID configurations. The system consistently maintained accurate lane following and effectively avoided obstacles, even under challenging night time conditions. The results show that computer vision techniques, when carefully integrated, can deliver reliable and computationally efficient autonomous driving behavior. OpenCV proved crucial for real time visual data processing, while the Tesla Model 3 robot in Webots provided a realistic vehicle model for simulation and testing. This work provides a solid foundation for future improvements, including dynamic obstacle handling, better performance under diverse lighting conditions, and refined avoidance strategies to enable safer and smoother navigation at higher speeds.**

I. Introduction

In recent years, autonomous driving has emerged as one of the most promising areas in Robotics and Artificial Intelligence, with increasing relevance in applications such as personal mobility, logistics, and urban transportation. By reducing human error and enhancing road safety, autonomous vehicles aim to revolutionize mobility systems. While many advances have been achieved under favorable daylight conditions, ensuring reliable performance at night remains a significant challenge, particularly for perception systems that rely on visual information. In this context, detecting road lanes and identifying obstacles in low light scenarios becomes crucial for safe autonomous navigation.

This research addresses the challenge of enabling an autonomous vehicle to operate in a simulated urban environment under night time conditions. The vehicle must be capable of detecting and following a yellow lane, while simultaneously identifying and avoiding obstacles along its path. The challenge lies in achieving robust lane detection and obstacle avoidance despite the difficulties posed by poor lighting conditions.

The motivation behind this research stems from the real world necessity for autonomous vehicles to function reliably at night, where lighting conditions can vary drastically. In this context, the central research question is: how can computer computer vision and control algorithms be combined to enable effective night time lane following and obstacle avoidance in autonomous vehicles? The hypothesis is that a pipeline based on HSV thresholding, Canny edge detection, Hough Transformation, and LiDAR-based obstacle detection, coupled with a PID controller for steering, can achieve stable and reliable behaviour in night time simulated environments.

The primary objective of this project is to develop and evaluate an autonomous control pipeline capable of detecting and following a yellow lane using computer vision techniques implemented with OpenCV. Steering control is managed by a PID controller that minimizes a composite heading error, calculated from both the angular deviation of the detected lane and its position relative to the center of the image. Additionally, the system incorporates LiDAR sensors for obstacle detection and dynamically adjusts the vehicle's trajectory to safely avoid any obstacles encountered along the path.

The remainder of this paper is organized as follows. Section II describes the methodological approach, including the simulation environment, sensor configuration, lane detection, control algorithms, and obstacle avoidance strategies. Section III presents the experimental evaluation of the developed system in simulated urban night conditions. Finally, Section IV summarizes the conclusions and discusses potential directions for future work.

II. Related Work

Lane detection and autonomous navigation have long relied on robust vision algorithms. Yu et al. [1] proposed a method combining the HSV color model with edge detection for

vehicle detection, illustrating the effectiveness of color-space filtering in real-time applications. For edge extraction, the Canny algorithm [3] remains a widely adopted technique due to its precision and noise reduction, and is often used in combination with the Hough Transform [4] for line detection, as demonstrated by Jahan et al. [2] in low-resource embedded systems like the Raspberry Pi.

At night, vision-based systems face additional challenges such as reduced contrast and increased noise. Coenen and Hermans [7] addressed these issues by proposing enhancements to camera-based lane detection under low-light conditions, supporting the relevance of vision-only pipelines during nighttime operation.

PID control is a standard strategy in vehicle steering systems for minimizing lateral and heading errors. Åström and Murray [5] provide foundational insights into the tuning and stability of feedback controllers for such dynamic systems.

LiDAR-based obstacle detection provides complementary spatial awareness, especially valuable in environments with limited visibility. Chen et al. [8] integrated sensor perception to enhance driving decisions through affordance learning.

For simulation and prototyping, Webots [6] offers a flexible and high-fidelity platform for developing and testing robotic control systems in structured scenarios, such as urban nighttime driving environments.

Finally, metrics such as lane error, heading error, error variance and number of oscillations remain key indicators for evaluating the performance of lane-following and control algorithms, as highlighted in broader surveys of planning and control techniques [9].

III. Methodological approach

A. Simulation Environment

The autonomous driving system was developed and tested using the **Webots** simulator.

B. Sensor Setup and Data Collection

The simulated autonomous vehicle was instrumented with two principal sensing modalities, each serving a distinct yet complementary role in the perception pipeline.

Camera: A forward-facing RGB camera with a planar projection was employed for visual lane detection. Initially configured with a resolution of 1024 x 512 pixels, a field of view of 1.0 radians, and a maximum sensing range of 5 meters, the camera was mounted with a translation vector of [0.72, 0, -0.5] and a rotation vector of [0, -1, -0.15], orienting it towards the road surface. During Experiment 1, multiple configurations were evaluated. The final setup, used throughout subsequent experiment, featured a reduced image resolution of 512 x 256 pixels, which provided an optimal balance between processing speed and accuracy for real-time lane detection. This sensor provided continuous visual input, which was processed using computer vision techniques to identify and track lane markings with high temporal resolution, enhancing robustness under varying lighting conditions.

LiDAR: A single-layer LiDAR sensor was mounted in a fixed forward-facing position (translation: [0,0,0]; rotation: [0,0,1,0]) to facilitate obstacle detection. The sensor operated with 45 horizontal rays across a horizontal field of view of 1.57 radians, and a vertical field of view of 0.1 radians. It offered a sensing range starting at 0.05 meters and extending up to 5 meters, enabling detection of nearby and mid-range obstacles directly in the vehicle's path.

Both sensors were activated at each simulation timestep, ensuring synchronized data acquisition aligned with the control loop frequency. This configuration allowed the system to maintain temporal coherence between perception and actuation, thereby supporting accurate lane following and timely obstacle avoidance within the constraints of the simulated urban environment.

C. Lane Detection and Image Processing

The lane detection pipeline processed raw camera images in real time using a sequence of image processing and geometric analysis steps implemented with OpenCV. The procedure is detailed as follows.

Image Capture and Preprocessing: Images were continuously acquired from the forward-facing RGB camera in `BGRA` (Blue, Green, Red, Alpha) format and subsequently converted to `BGR` (Blue, Green, Red). To enhance robustness against varying illumination, each frame was transformed into the HSV color space, which allows more effective color-based segmentation.

Color Segmentation: A yellow color mask was applied by thresholding the HSV image within the range `[20, 100, 100]` to `[35, 255, 255]`. This operation was constrained to the lower 30% of the image height, focusing the analysis on the road surface and mitigating the influence of irrelevant background elements.

Edge Detection: The masked image region was processed with the Canny edge detector using lower and upper thresholds of 100 and 200, respectively. This step emphasized the contours of the lane boundaries, facilitating subsequent line extraction.

Line Detection and Selection: A Probabilistic Hough Line Transform was applied to the edge detected image with the following parameters: resolution $\rho = 1$, angle resolution $\theta = \pi/180$, minimum line length of 30 pixels, maximum gap of 10 pixels, and a detection threshold of 50. All detected lines were rendered for debugging, and the longest line was selected as the primary candidate for heading estimation.

Steering Error Estimation: For the selected line, angular error was computed as the deviation between its orientation and a vertical reference ($-\pi/2$ radians). A lateral error was also calculated based on the horizontal distance between the line's midpoint and the image center. A combined steering error metric was produced by weighting the angular and po-

sitional errors, with angular deviations given higher influence when the line was significantly inclined.

Fallback Strategy: In scenarios where no valid lines were detected by the Hough Transform, a fallback mechanism based on region moments was employed. The centroid of the yellow lane region was computed from the binary mask, and its offset from the image center was used to estimate lateral deviation. If this deviation fell within a predefined tolerance, the system maintained the current trajectory by setting the steering error to zero, ensuring smooth continuity in navigation.
The following images illustrate the intermediate stages of the lane detection pipeline, showcasing how raw input from the camera is progressively transformed through color segmentation, edge detection, and line extraction processes to estimate the vehicle's heading and lateral position.

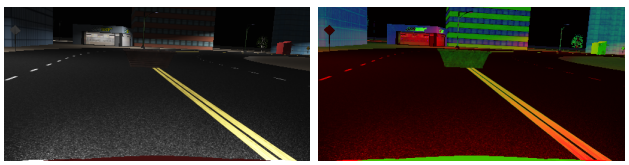

Fig. 1: (Left) Original RGB frame,
(Right) HSV yellow mask.

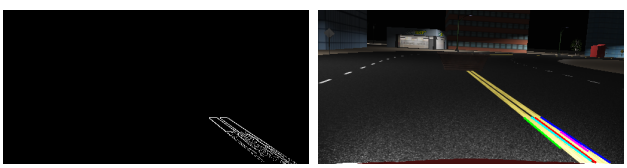

Fig. 2: (Left) Canny edges,
(Right) Hough line detection.

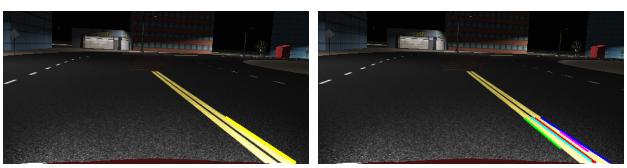

Fig. 3: Selection of best line

D. Control Algorithm

A **PID controller** was employed to mitigate steering deviations detected by the lane detection system. The controller parameters were empirically tuned to achieve an optimal balance between responsiveness and stability. To prevent abrupt or unsafe steering actions, the control output was bounded within permissible limits.

The PID controller computes the control input $u(t)$ based on the error signal $e(t)$ according to the following expression:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt}$$

where:

- K_p is the proportional gain, responsible for correcting the present error;
- K_i is the integral gain, which eliminates steady-state error by considering the accumulation of past errors;
- K_d is the derivative gain, which predicts future error based on its rate of change, improving system stability.

Vehicle speed was modulated based on steering error magnitude, with a baseline cruising speed adjusted downward during sharp turns to maintain safety and control.

E. Obstacle Detection and Avoidance

Obstacle detection was performed using filtered LiDAR range data. When an object was identified within a threshold distance of 7 meters, the system initiated an avoidance protocol. This protocol involved dynamically reducing the vehicle's speed in proportion to the obstacle's proximity, while concurrently adjusting the steering angle to navigate around it. The direction of avoidance was determined by comparing lateral LiDAR readings, selecting the path offering the greatest clearance. Steering corrections were applied incrementally to ensure smooth maneuvering and prevent abrupt changes in trajectory. The avoidance actions continued for a short buffer period after the obstacle was no longer detected, followed by a gradual transition back to the nominal lane following control.

IV. Experimental Evaluation

A. Test Scenarios

Two distinct maps were designed to evaluate the robot's behavior in different driving contexts:

Road A (Experiment 1): This scenario featured a continuous yellow lane on a curved road with no obstacles. It was used to assess the lane following capabilities of the robot under controlled conditions. Various parameters such as PID gains, vehicle speed, and camera resolution were systematically varied to identify the best configuration. Results from these tests are reported later in the form of comparative tables.

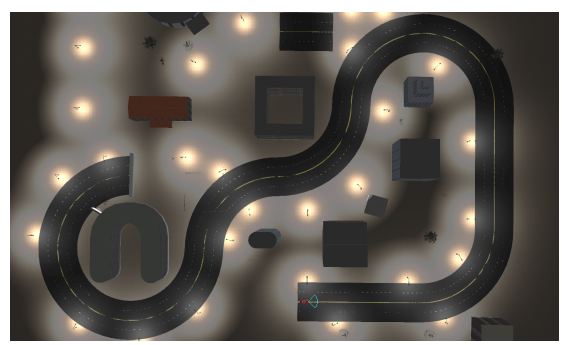

Fig. 4: Road A used in Experiment 1

Road B (Experiment 2): This scenario introduced a more complex road layout with multiple turns and static obstacles placed along the lane. While the simulated environment remained the same (*City Night*), the increased difficulty required

the robot to perform both lane-following and obstacle avoidance simultaneously. Due to the higher number of obstacles, the vehicle's speed had to be reduced, and new PID and vision settings were tested, as the configuration from Experiment 1 was insufficient for stable navigation in this more demanding environment.

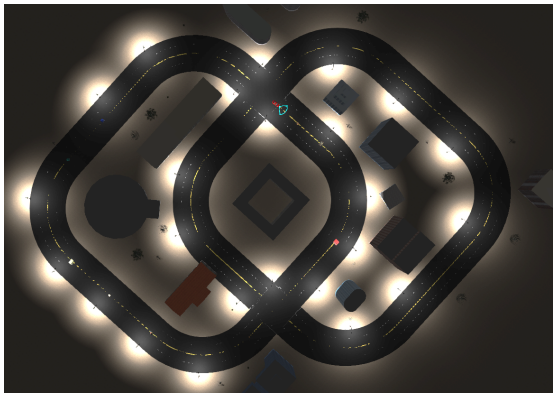

Fig. 5: Road B used in Experiment 2

B. Evaluation Methodology

The performance of the autonomous vehicle was evaluated using four quantitative metrics, designed to assess both the accuracy and the stability of the robot's behavior during the simulations:

1) **Lane Error (L. E.)**
This metric represents the robot's combined lateral and angular deviation from the yellow lane. It was calculated by comparing the position and orientation of the detected lane line in the camera image with the center of the image frame. Lower values indicate that the robot remained properly aligned and centered with the lane, while higher values suggest misalignment or drift in position and heading.
2) **Error Variance (E. Var.)**
To assess tracking stability, the variance of the lane error was computed across the duration of each simulation. A low variance value implies that the robot maintained a consistent position relative to the lane, whereas a high variance indicates frequent deviations and unstable tracking.
3) **Heading Error (H. E.)**
This metric measures the angular deviation between the robot's heading and the orientation of the lane. It was obtained by comparing the detected angle of the lane with the robot's forward direction, combined with a weighted lateral offset from the lane center to capture overall alignment. Values close to zero indicate proper heading and positioning, while larger values suggest misalignment or poor orientation during navigation.
4) **Oscillations (Osc)**
Oscillations refer to the number of abrupt steering direction changes observed during the simulation. This was quantified by counting significant changes in the steering angle over time. A lower number of oscillations reflects smoother and more stable control behavior, while excessive oscillations indicate an unstable or overly sensitive control system.

To assess the control system, multiple runs were conducted with varying combinations of PID gains and speed settings. These configurations were evaluated during Experiment 1 to identify effective parameter ranges. Performance was measured based on metrics such as lateral stability, trajectory smoothness, and successful completion of the course. Although the best performing configuration from this phase was selected for use in Experiment 2, the speed setting from Experiment 1 proved suboptimal under the new conditions. As a result, different speed values were tested in Experiment 2 to account for the changed scenario.

The following tables summarize the tested configurations along with their corresponding performance results, organized by metric:

TABLE I: Performance metrics for each PID configuration tested on Experiment 1

PID (Kp, Ki, Kd)	L. E.	E. Var.	H. E.	Osc
(0.04, 0.001, 0.01)	7.454	128.452	0.778	1538
(0.04, 0.000, 0.01)	15.586	140.569	0.987	663
(0.04, 0.005, 0.01)	6.885	109.829	0.767	1490
(0.04, 0.001, 0.01)	35.078	163.767	0.151	101
(0.00, 0.001, 0.01)	87.544	853.452	1.016	451
(0.08, 0.001, 0.01)	10.515	199.907	1.027	599
(0.04, 0.001, 0.00)	5.577	223.798	1.085	30
(0.04, 0.001, 0.04)	10.580	305.220	0.822	1748

TABLE II: Performance metrics for each Camera Resolution tested on Experiment 1

Resolution (W x H)	L. E.	E. Var.	H. E.	Osc
512 x 256	7.454	128.452	0.778	1538
2048 x 1024	193.281	44590.596	1.151	387
1024 x 512	218.216	13187.874	0.939	262

TABLE III: Performance metrics for each speed tested on Experiment 1

Speed (km/h)	L. E.	E. Var.	H. E.	Osc
45	6.470	84.541	0.729	1860
50	7.210	119.702	0.752	1754
55	7.454	128.452	0.778	1538
60	10.055	179.999	0.988	1129

TABLE IV: Performance metrics for each speed tested on Experiment 2

Speed (km/h)	L. E.	E. Var.	H. E.	Osc
15	75.453	4335.755	0.779	855
20	7.837	1496.902	0.552	55
25	203.773	2652.229	0.414	142
30	0.061	4.372	0.068	25

V. Analysis of Results

The experimental evaluation provided key insights into how PID parameters, camera resolution, and vehicle speed affect the autonomous system's lane-following and obstacle avoidance capabilities.

a) PID Tuning and Control Behavior:

In Experiment 1, multiple PID configurations were tested to identify the most stable and responsive setup. As shown in Table 1, the configuration $(K_p = 0.04, K_i = 0.001, K_d = 0.01)$ offered a good balance between accuracy and stability, achieving a relatively low lane error (7.454) and heading error (0.778), while maintaining controlled oscillations (1538). This setup was chosen as the baseline due to its consistent performance across all metrics.

Eliminating the integral term (e.g., $K_i = 0.000$) or increasing the derivative gain excessively resulted in instability, higher heading errors, or excessive oscillations. Similarly, configurations with zero proportional gain caused severe lane deviation and loss of tracking. These findings highlight the importance of balanced tuning of all PID components for smooth yet effective control.

b) Resolution and Real-Time Responsiveness:

Table 2 evaluates the impact of camera resolution on control performance. The 512×256 resolution was selected as optimal, providing a favorable trade-off between accuracy and computational load. Higher resolutions such as 1024×512 and 2048×1024 caused significant performance degradation, especially in terms of error variance, likely due to increased processing delays that compromised real-time responsiveness. Consequently, the lower resolution was adopted for subsequent experiments.

c) Speed and Control Limitations:

As summarized in Table 3, vehicle speed directly influenced lane tracking accuracy. While 45 km/h yielded the lowest lane error and acceptable stability, 55 km/h offered the most balanced results when combined with the chosen PID settings and camera resolution. Thus, 55 km/h was set as the default speed in Experiment 1.

However, the system's ability to follow lanes adequately at 55 km/h in open-road conditions was not sufficient for more complex scenarios involving obstacle avoidance.

d) Motivation and Results of Experiment 2:

The motivation for Experiment 2 was to apply the best-performing parameters from Experiment 1 in a different scenario featuring a more complex road layout with static obstacles. However, the optimal speed used previously could not be maintained because the robot struggled to stay on track at higher speeds in this new environment. For these reasons, several lower speeds were tested (Table 4) to find a suitable balance between control and obstacle avoidance.

Although the results at 30 km/h initially suggested near-perfect lane tracking—with minimal lane error (0.061), low heading deviation (0.068), and very few oscillations (25), this performance was misleading. After the first obstacle, the vehicle failed to stay on course and exited the route entirely. As a result, these low values reflect only the initial moments of the run and are not representative of sustained performance, highlighting the configuration's overall inadequacy for this scenario. For a balance between real-time responsiveness and effective obstacle avoidance, 20 km/h was selected as the optimal operating speed. This setting maintained stable control (heading error 0.552, oscillations 55) while allowing the vehicle to successfully detect and avoid obstacles.

e) Summary: Overall, moderate PID gains combined with reduced speeds enabled the most stable and successful navigation in both experimental scenarios. Higher speeds increased risks of overshooting curves or failing to avoid obstacles, especially in the presence of static obstacles (Road B). The inclusion of the derivative term was essential for maintaining directional stability, particularly when responding to sudden trajectory changes during obstacle avoidance.

VI. Conclusions and future work

This work explored the use of computer vision techniques implemented with OpenCV, combined with LiDAR sensor data, to enable visual perception and obstacle detection in a simulated urban environment under night time conditions. The results obtained throughout the project highlight the effectiveness and limitations of control methods, particularly the PID controller, when applied to real time navigation tasks.

In particular, the PID controller, once carefully tuned, significantly enhanced the stability and responsiveness of the vehicle during lane following. By adjusting the control gains, the system was able to maintain alignment with the yellow lane line with minimal oscillation, ensuring a smooth and consistent trajectory. This outcome demonstrates the practicality of PID control in structured environments where the dynamics remain relatively predictable. Moreover, the use of the HSV color space for lane detection proved to be robust across a range of lighting conditions. The segmentation technique consistently and reliably isolated the yellow lane markings, which in turn enabled accurate lateral error estimation for the controller. As a result, this method demonstrated strong suitability for real time navigation, particularly in night time scenarios.

Furthermore, experimental analysis revealed important trade-offs between vehicle speed and control performance. At moderate speeds, particularly around 50 km/h, the PID controller achieved a favorable balance between responsiveness and stability. However, as speed increased, the system exhibited greater tracking errors and reduced heading stability, highlighting its limitations under more dynamic conditions.

In contrast, obstacle avoidance required considerably lower speeds, around 20 km/h, in order to ensure safe and reliable maneuvering. These findings underscore the importance of reducing speed in safety-critical situations where timely perception and reaction are essential.

Taken together, these results point to several broader insights. Autonomous navigation performance is highly sensitive to vehicle speed, environmental complexity, and the effectiveness of sensor integration. While the PID controller offers a solid baseline for structured tasks, its performance diminishes in faster or more unpredictable scenarios.

Looking ahead, several promising directions can extend this project. A key area is the integration of dynamic obstacle handling, which would enable the system to detect and respond not only to static objects but also to moving vehicles and pedestrians. Achieving this will require enhanced perception capabilities and adaptive planning strategies to safely navigate complex, real-world scenarios.

Additionally, testing under a wider range of weather and lighting conditions would help evaluate the system's robustness more comprehensively. Such conditions are likely to challenge the current vision pipeline based on HSV thresholding and may call for improved algorithms capable of maintaining reliable lane detection despite environmental variability.

Moreover, the obstacle avoidance mechanism itself could be refined to improve path planning and the smoothness of steering maneuvers. By optimizing turning behaviors and route selection during obstacle negotiation, the system could offer safer and more comfortable control, especially in cluttered or unpredictable environments. These improvements will be critical for enabling the vehicle to operate reliably at higher speeds while maintaining safety and stability.

Overall, these enhancements would contribute significantly to increasing the generalization, reliability, and real world applicability of autonomous navigation systems, particularly in unstructured and dynamic urban settings.

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